

Mohammad Madadi

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EDUCATION

Sharif University of Technology

B.Sc. in Mechanical Engineering

Tehran, Iran

Sep 2019 – Jul 2024

- GPA: 17.66/20 (3.77/4.0)
- Note: Excluding 9th semester (medical leave), GPA is 18.13/20 (3.88/4.0).

RESEARCH INTERESTS

- Robotics & Autonomous Systems
- Embodied AI

PUBLICATIONS

- **"The Role of Deep Reinforcement Learning in Motion Planning for Robotic Arm Assistive Systems."**
International Conference of Iranian Society of Mechanical Engineers (ISME), 2025.

RESEARCH EXPERIENCE

University of Hong Kong (HKU)

Research Intern, Agentic Intelligence Lab

Hong Kong

Sep 2025 – Present

- Supervisor: Prof. Jiayu Chen
- Focus: Continual Reinforcement Learning for Humanoid Robotic Control.

Sharif University of Technology

Research Assistant, Dept. of Mechanical Engineering

Tehran, Iran

Jul 2024 – Dec 2024

- Supervisor: Prof. Mohammad Taghi Ahmadian
- Focus: Deep Reinforcement Learning in Motion Planning for Robotic Arm Assistive Systems.

Sharif University of Technology

Research Assistant, Mechatronics Lab

Tehran, Iran

Sep 2023 – Jul 2024

- Supervisor: Prof. Gholamreza Vosoughi
- Focus: Developing "Sharif Exoskeleton" for paralyzed patients.

Sharif University of Technology

Research Assistant, Spine Biomechanics Lab

Tehran, Iran

Sep 2022 – Jul 2023

- Supervisor: Prof. Navid Arjmand
- B.Sc. Thesis: Risk assessment of workers performing seated tasks and implementing interventions to reduce spinal injury risks utilizing ergonomic and biomechanical tools.

Research Center of Intelligent Neuro-Rehabilitation

Research Assistant

Tehran, Iran

Sep 2022 – Feb 2023

- Supervisor: Prof. Navid Arjmand
- Collected In-Vivo body movement data for motion and force analysis in gait and manual material handling activities.

SELECTED PROJECTS

- **Maze Solver using Markov Decision Processes (RL):** Implemented RL algorithms to solve maze environments.
- **Control & Trajectory Planning for 3 DOF Robot:** Designed control schemes and path planning for a robotic manipulator.
- **Action Recognition System:** Developed a human action recognition system using Arduino and IMU sensors; analyzed data using Neural Networks in Python.
- **Microswimmer Dynamics:** Derived dynamic equations for a 4-sphere microswimmer operating in 2D environments.
- **Intelligent Security System:** Designed and implemented an electronic entrance door security system.

TEACHING EXPERIENCE

- **Teaching Assistant**, Machine Dynamics (Prof. Saeed Behzadipor), Sharif Univ. of Tech.
- **Teaching Assistant**, Mechanical Engineering Graphics (Prof. Hamid Haghshenas), Sharif Univ. of Tech.

HONORS & AWARDS

- **First Class Honours**, B.Sc. in Mechanical Engineering, Sharif University of Technology 2024
- Ranked 9th in the National University Entrance Exam (Konkur) among 164,000 participants 2019

SKILLS & LANGUAGES

- **Programming:** Python, C, MATLAB/Simulink, Arduino.
- **Frameworks:** PyTorch.
- **Engineering Software:** Ansys Fluent, SOLIDWORKS, 3dspp, Jack, Vicon Nexus, AnyBody.
- **Languages:** English (TOEFL: 108/120), Persian (Native).